

ROCK OF AGES, MI, USA

WMO: 994130

Lat: 47.867N

Lon: 89.313W

Elev: 602

StdP: 14.38

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -9.8	(c) -4.6	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 46.8	(k) 14.7	(l) 42.5	(m) 18.8	(n) 19.2	(o) 340	(p) 0.925

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 8.5	(c) 73.2	(d) N/A	(e) 70.7	(f) N/A	(g) 68.4	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 17.1	(p) 270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 41.1	(b) 36.2	(c) 32.9	(d) DB	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A	(q) N/A
(4) 41.1	(5) 36.2	(6) 32.9	(7) WB	(8) N/A	(9) N/A	(10) N/A	(11) N/A	(12) N/A	(13) N/A	(14) N/A	(15) N/A	(16) N/A	(17) N/A	(18) N/A	(19) N/A	(20) N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	40.1	15.2	17.3	27.7	37.2	45.4	53.3	60.4	61.9	57.0	45.9	34.2	24.4
	DBStd	17.64	12.27	10.55	9.79	7.24	5.56	4.59	5.69	4.73	5.82	6.46	8.75	9.42
	HDD50	4696	1080	916	693	385	166	18	1	0	11	159	473	794
	HDD65	9139	1545	1336	1158	834	609	352	164	121	246	592	923	1259
	CDD50	1080	0	0	0	0	22	116	322	368	220	32	0	0
	CDD65	47	0	0	0	0	0	0	20	23	4	0	0	0
	CDH74	58	0	0	0	0	0	9	33	15	1	0	0	0
	CDH80	3	0	0	0	0	0	0	3	0	0	0	0	0
Wind	WSAvg	17.1	20.6	18.7	17.1	16.4	15.8	13.8	13.8	14.5	16.0	18.8	19.6	20.1
Precipitation	PrecAvg	28.8	1.9	1.2	1.5	2.5	2.9	3.3	3.0	2.7	3.2	3.1	2.4	1.9
	PrecMax	35.5	3.7	2.3	2.8	7.4	4.4	5.9	5.5	4.5	6.6	7.9	4.5	3.1
	PrecMin	23.3	0.7	0.6	0.6	0.8	1.6	0.9	1.0	1.1	1.2	1.2	0.9	0.6
	PrecStd	3.1	0.7	0.4	0.6	1.4	1.0	1.3	1.3	1.0	1.3	1.6	0.9	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.5	40.9	50.6	57.1	67.6	74.6	78.2	75.9	72.5	64.4	51.5	41.8
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	37.0	35.9	45.1	52.5	60.8	68.2	73.4	72.8	68.6	60.1	48.4	39.8
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	34.9	33.2	41.0	49.2	57.3	64.5	70.8	70.4	66.3	56.9	46.6	38.2
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	32.3	30.9	38.0	46.3	54.1	61.6	68.3	68.3	64.1	53.7	45.0	36.8
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Mean Daily Temperature Range	5% DB	MDBR	11.4	10.8	9.7	10.0	11.8	14.8	12.8	8.5	7.8	7.1	7.8	9.5
		MCDBR	10.6	9.6	11.8	15.1	18.1	20.1	15.2	11.1	10.7	10.9	7.5	7.9
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Clear-Sky Solar Irradiance	taub	0.267	0.279	0.305	0.343	0.366	0.384	0.396	0.378	0.360	0.322	0.294	0.271	
	taud	2.471	2.417	2.398	2.365	2.398	2.368	2.400	2.457	2.480	2.541	2.532	2.493	
	Ebn at Noon	264	284	292	288	285	279	273	274	269	265	251	248	
	Edh at Noon	20	26	31	36	36	37	36	32	29	23	19	17	
All-Sky Solar Radiation	RadAvg	317	626	1079	1491	1750	1904	1925	1653	1189	702	355	244	
	RadStd	36	72	101	162	117	128	94	117	72	85	46	27	

Historical Trends

	DBAvg	Heating		Cooling		Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (21 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page